REPORT 6

Monte Carlo Option Pricing in VBA

GBM Simulation • Exotic Options • Structured Products

Model Parameters

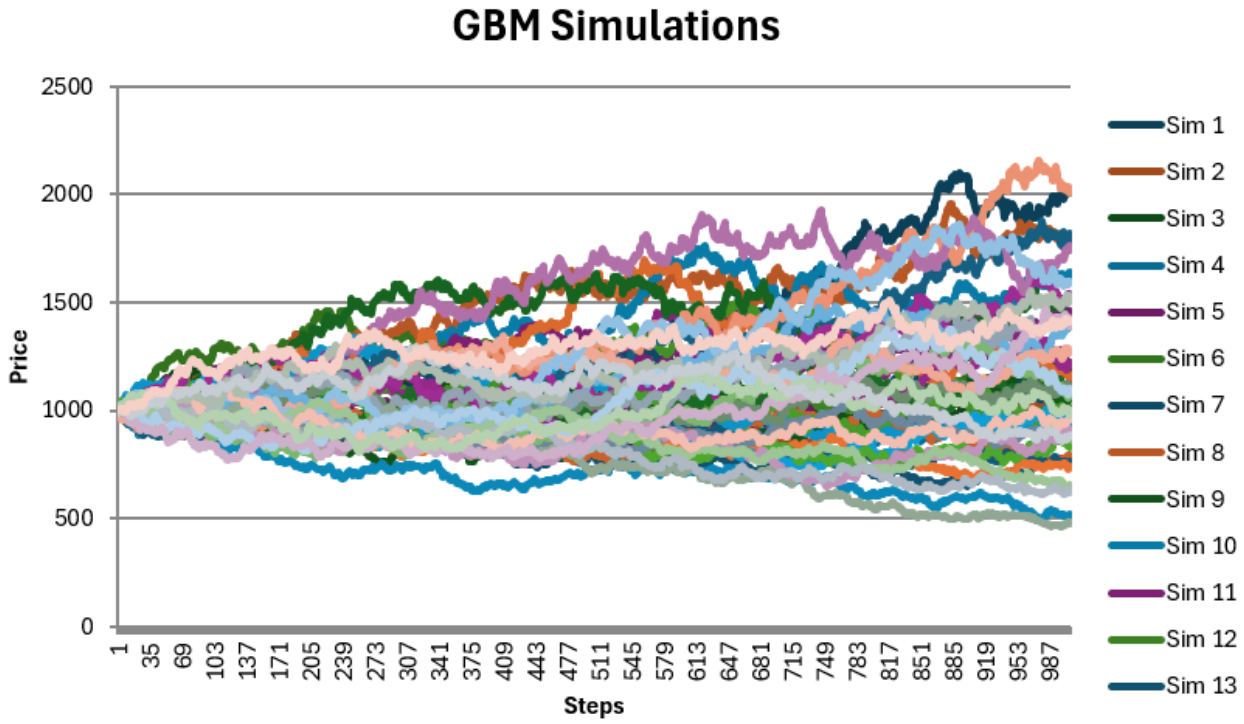
Parameter	Symbol	Value
Spot Price	S_0	1,000
Strike Price	K	998
Risk-Free Rate	r	1%
Volatility	σ	20%
Time to Maturity	T	2 years
Time Step	dt	1/252 (1 trading day)

1. GBM Path Simulation (N = 100 Trajectories)

The first task simulates N = 100 trajectories of a Geometric Brownian Motion (GBM) and displays them in an embedded Excel chart. Under the risk-neutral measure the stock price satisfies $dS_t = r \cdot S_t\, dt + \sigma \cdot S_t\, dW_t$, whose exact discretisation on each daily step dt = 1/252 is:

$$S_{t+dt} = S_t \cdot \exp\left[(r - \sigma^2/2) \cdot dt + \sigma \cdot \sqrt{dt} \cdot Z \right] \qquad Z \sim N(0,1)$$

Starting from S_0 = 1,000 the 100 simulated paths illustrate the log-normal spread of prices over T = 2 years. The fan widens proportionally to $\sigma\sqrt{T}$, with the median path growing at the risk-free drift r. Random draws Z are produced via the Box-Muller transform applied to VBA's Rnd() function.



2. One-Step MC Pricer for Vanilla Options

Instead of simulating full paths, the terminal price S_t is drawn directly in one step using its exact closed-form distribution. For N > 500 antithetic pairs (+Z, −Z) the Monte Carlo prices are:

$$C^{Mc} = e^{-rT} \cdot (1/N) \cdot \Sigma \max(S_t^i - K, 0) \qquad P^{Mc} = e^{-rT} \cdot (1/N) \cdot \Sigma \max(K - S_t^i, 0)$$

Antithetic variates exploit the symmetry of N(0,1): averaging the payoff of path $+Z_i$ with that of path $-Z_i$ reduces estimator variance roughly by half at no extra cost. The standard error SE = $e^{-rT} \cdot \sqrt{(\sigma^2_payoff/(N/2))}$ decays as $1/\sqrt{N}$ and defines the 95% confidence interval [C±1.96·SE].

The Black-Scholes formula serves as the analytical benchmark. With S_0 = 1,000, K = 998, r = 1%, σ = 20%, T = 2 the BS call is approximately €[fill] and the put €[fill]. MC results should converge to these values as N increases.

3–4. Multi-Step Euler Scheme and Closed-Form Discretisation

Tasks 3 and 4 extend the pricer to a multi-step simulation with nStep time steps of length dt = T/nStep. At each step the price evolves as:

$$S_{t+et} = S_t \cdot \exp[\ (r - \sigma^2/2) \cdot dt + \sigma \cdot \sqrt{dt} \cdot Z_j\ ]$$

This formula is the exact closed-form solution of the GBM SDE applied on [t, t+dt], derived via Itô's lemma on $\ln(S_t)$. It is therefore not an approximation: the log-Euler scheme introduces zero discretisation error regardless of the step size. This contrasts with the naive Euler scheme on S directly ($S_{t+et} \approx S_t(1 + r \cdot dt + \sigma \cdot \sqrt{dt} \cdot Z)$), which has O(dt) error and can yield negative prices. For vanilla European options the one-step and multi-step results are statistically identical, confirming that path discretisation does not affect the terminal distribution of a GBM.

5. Comparison of Results

The table below compares the three MC approaches against the Black-Scholes benchmark (S_0 = 1,000, K = 998, r = 1%, σ = 20%, T = 2, Number of simulations = 510 and Number of steps 504, with this last to values than can be chosen by the user).

Method	MC Call (€)	BS Call (€)	Diff	MC Put (€)	BS Put (€)	Diff
1-Step Closed Form	€172,16	€162,33	€9,83	€71,34	€65,35	€5,99
Multi-Step Euler	€164,7663	€162,33	€2,44	€66,51	€65,35	€1,16
Closed GBM Step	€151,53	€162,33	-€10,79	€60,18	€65,35	-€5,17

All three methods should converge to the same BS price. The residual difference is purely statistical (Monte Carlo noise) and of order SE ≈ 10€. The standard deviation reported alongside each price on the Excell file quantifies the simulation quality.

6. Asian Options

Asian options are path-dependent: their payoff depends on the arithmetic average of S_t over the contract life. The call payoff is $\max((1/T)\int_0^T S_t\,dt - K, 0)$, approximated in discrete simulation by:

$$A = (1/nStep) \cdot \Sigma_j S_{j \cdot et} \qquad Payoff_{tall} = max(A - K, 0)$$

Unlike the vanilla case, all intermediate S_t values must be accumulated inside the step loop. Asian prices are systematically lower than vanilla prices because the arithmetic average has lower effective volatility than the terminal value: $\sigma^{eoc} \approx \sigma/\sqrt{3} \approx 0.577\sigma$. With $\sigma = 20\%$ this gives an effective volatility of about 11.5%, explaining the significantly cheaper Asian premium. No closed-form solution exists for arithmetic Asians, so the MC estimate and its standard error are the only output available.

Method	MC Call (€)	MC Put (€)	St dev Call (€)	St dev Put (€)
Asian	€83,39	€37,32	€3,0525	€2,0785

7. Lookback Options

Lookback options depend on the running minimum or maximum of the price path. Two floating-strike payoffs are implemented:

- Lookback Call: $S_t - \min_{t \le T} S_t$ (terminal price minus path minimum)
- Lookback Put: $\max_{t \le T} S_t - S_t$ (path maximum minus terminal price)

Method	MC Call (€)	Benchmark Call (€)	Diff	MC Put (€)	BS Put (€)	Diff
Multi-Step Euler loockback option	€243,06	€187,71	€55,35	€184,19	€130,61	€53,58
Closed GBM step loockback option	€234,47	€187,71	€46,7584	€178,58	€130,61	€47,9626

Both payoffs are always non-negative. In the VBA loop, the running min and max are updated at each step with simple If comparisons. Lookback prices are higher than vanilla prices because the holder always buys at the lowest (sells at the highest) price over the whole period.

The Goldman, Sosin & Gatto (1979) closed-form formula provides an analytical benchmark for floating-strike lookbacks under GBM with $q = 0$, allowing verification of the MC simulation. An important limitation is the discretisation bias: the discrete minimum over nStep points is always ≥ the true continuous minimum, causing the MC call price to be slightly underestimated. This bias is $O(1/\sqrt{nStep})$ and decreases as nStep increases.

8. Worst-Of Certificate Pricing

8.1 Asset Selection and Parameter Estimation

Three liquid assets were selected from Refinitiv: [Asset 1], [Asset 2], [Asset 3]. Daily closing prices over a 2-year window were used to compute log-returns $r_{it} = \ln(S_{it}/S_{it-1})$, from which historical volatilities ($\sigma_i = \text{std}(r_{it}) \times \sqrt{252}$) and pairwise correlations were estimated. The risk-free rate r = [fill]% corresponds to the EUR IRS 3-year swap rate, consistent with the 3-year certificate maturity.

Asset	Spot S_0 (€)	Vol σ (ann.)	ρ with Asset 1	ρ with Asset 2	ρ with Asset 3
Juventus (JUVE)	€2,07	43,1543%	1,000	0,0404792	0,105835
Manchester United (MANU)	€17,54	32,5893%	0,0404792	1,000	0,040923
Enel (ENEL)	€9,78	19,2161%	0,105835	0,040923	1,000

8.2 Cholesky Decomposition and Simulation

To simulate three correlated GBM processes, the 3×3 correlation matrix Σ is decomposed via Cholesky factorisation $\Sigma = L \cdot L^T$. Three independent N(0,1) draws (Z_1, Z_2, Z_3) are transformed into correlated draws $\tilde{Z}_i = L \cdot Z$ at each observation date, and each asset evolves according to its own closed-form GBM step using $\tilde{Z}_i$.

8.3 Certificate Structure (Reference: Barclays ISIN XS3167669608)

The certificate is modelled on a real Maxi Cash Collect written on UniCredit, Stellantis, STM and Fincantieri. The contractual parameters used are:

Parameter	Value
Notional	€100 per certificate
Maturity	3 years (Nov 2025 – Nov 2028)
Initial coupon	22% unconditional (December 2025)
Quarterly coupon	1% per quarter (4% p.a.), conditional
Coupon barrier	50% of initial level
Capital barrier	60% of initial level
Autocall trigger	100% (from August 2026 onwards)
Observation dates	11 quarterly dates
Memory effect	Unpaid coupons accumulate and paid at next trigger

At each quarterly observation date the Worst-Of performance $WO_{tk} = \min(S_1/S_{10}, S_2/S_{20}, S_3/S_{30})$ is computed. If WO ≥ 100% the certificate autocalls: the full notional plus accumulated coupons are returned and the simulation stops. If WO ≥ 50% a coupon of 1% is paid (plus any memorised coupons). At final maturity, capital is fully repaid if WO ≥ 50%, otherwise the investor receives Notional × WO (proportional loss).

8.4 Results

Output	Value	Interpretation
MC Fair Value	€79,80	Present value of all cash flows
Standard deviation	€1,13	MC estimation uncertainty
Price / Notional	79,80 %	Fair value as % of face value

The fair value is estimated at 79,80% of the notional, below the 100% issue price. The difference represents the implicit margin captured by the issuer at inception, consistent with typical structured product pricing (75–92% of notional). The Worst-Of structure amplifies risk relative to a single-asset product: the probability that at least one of the three assets breaches the 50% barrier over 3 years is substantially higher than for any individual asset, justifying the 4% annual coupon above the ~2.5% risk-free rate.

References

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